NgTrDat@cv-terminal:~

NgTrDat Wed 8 Oct - 17:05 ~ fastfetch -c $HOME/.config/fastfetch/CV.jsonc



NgTrDat@cv-terminal

Name : Nguyễn Trọng Đạt
Age-Gender : 22 - Male
Role : Computer Engineering
Study : Ho Chi Minh University of Technology
Focus : Robotics • AI • Drones • Embedded Systems
Location : Ho Chi Minh City, Vietnam
Email : trongdat4work@gmail.com
Languages : C/C++, Python, Shell script
Tools : VSCode, Git, ROS2, Gazebo
English : TOEIC L&R 970/990

NgTrDat Wed 8 Oct - 17:06 ~ ./script/ShowMyCV.sh

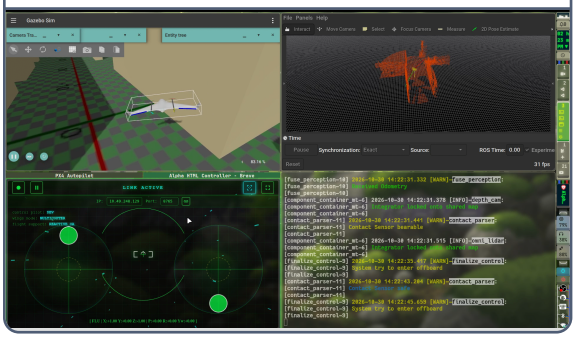
Professional_objective.txt

•Computer Engineering student focused on robotics, embedded systems, and autonomous drone software. Experienced with ROS2, PX4, Gazebo, and C++ development for real-time systems and simulation environments.

•Experienced in collaborative development environments using Git; proficient at reading, understanding, and contributing safely to shared codebases.

•Seeking a part-time engineering role, potentially transitioning to full-time after graduation.

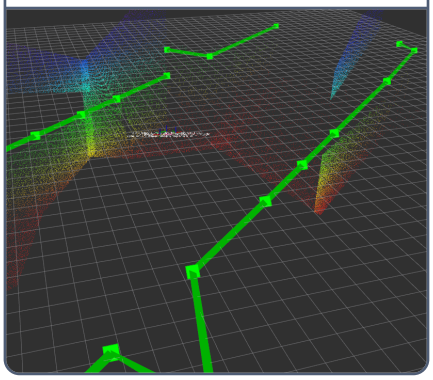
Drone_Simulation_Preview.png



Education.log

[Study] Ho Chi Minh University of Technology (BKU)
[Major] Bachelor of Engineering in Computer Engineering (Expected graduation 2027)
[GPA] 2.9/4.0
[Core Coursework] • Embedded System • Operating System • Machine Learning
• Microprocessors-Microcontrollers
• IoT-Oriented Logic Design Project
• AI-Oriented Multidisciplinary Project
[Passion Project] Developed a ROS2 and PX4 drone simulation pipeline in Gazebo to train AI models for VTOL acrobatic flight.

LIDAR_depth_scan.png



Skills.md

```
Programming:
  • Most proficient with C++, Python and Rust
  • Familiar with Java, Go, Zig
Robotics & Middleware ecosystem:
  • ROS2 (Jazzy)    • Gazebo Harmonic
  • PX4 Autopilot   • MicroXRCE-DDS
AI, Machine Learning and Data Engineering:
  • Latent Diffusion Learning    • Robotic data processing
  • Reinforcement Learning    • AI Edge Computing
Development tools: • VScode    • Git    • Linux OS
```

Contacts.txt

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Email : trongdat4work@gmail.com
Github: github.com/LemonKronos
LinkIn: www.linkedin.com/in/trong-dat
* All source code below is available on GitHub

About_me.txt

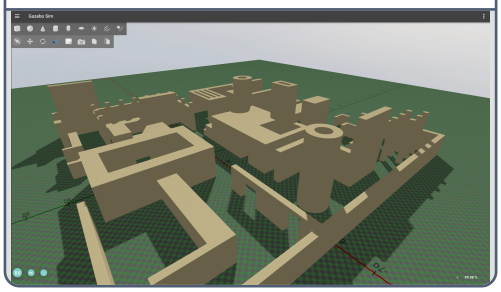
- I am a Drone enthusiast
- My goal is to become a Robotics Engineer
- Open-source mindset, member of Linux community
- Strong technical communicator, ready to collaborate in international teams

Multidisciplinary_Project_AI_Oriented.md

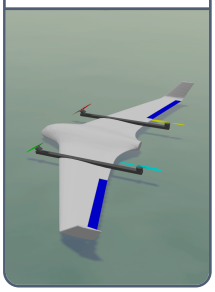
Project: Full-Pipeline Acrobatic Drone based on Latent Imitation Learning

- **Latent Imitation Learning:** Trains an AI model to imitate expert pilot behavior using latent space representations and performance-based evaluation, enabling policies that can potentially surpass the original demonstrations.
- **Goal:** framework designed to act as an "AI Co-pilot," denoising erratic, suboptimal human inputs into safe, expert-level 3D trajectories.
- **Main components:** Gazebo Harmonic for simulation and physics, PX4 Autopilot for high-frequency flight control (400Hz), Micro XRCE-DDS for PX4-ROS2 communication, and ROS2 as the offboard middleware handling sensors, messaging, and AI integration (80Hz). The acrobatic AI runs as a Python ROS2 node at a lower control frequency (0.5Hz).
- **Principle:** Inputs include raw human control commands, telemetry data, and environmental perception data (vision and 3D sensing). The AI generates an intended trajectory through the environment, which is validated by a safety and obstacle-avoidance layer before being executed by PX4 and translated into motor commands.

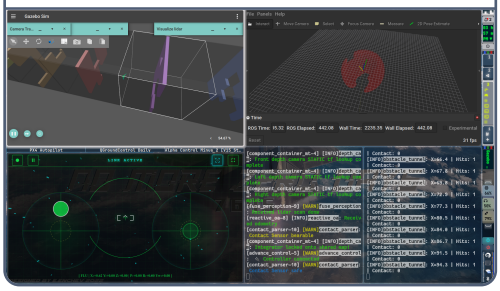
Test_Control_World.png



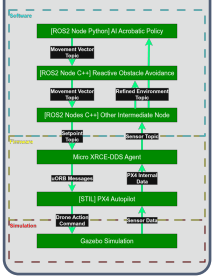
VTOL.png



Test_non-AI_behavior.png



Pipeline.png



Multidisciplinary_Project_AI_Oriented.md

- **Execution:** Collected expert control inputs and telemetry data in Gazebo to build a training dataset. The simulation environment consisted of a procedurally generated obstacle-tunnel world. Models were trained offline and evaluated through both offline testing and real-time user-controlled flight scenarios.
- **Results:** Successfully validated the non-AI control pipeline, including low-level obstacle avoidance, achieving autonomous flights of up to 1 km without collisions. However, the AI-controlled system produced noisy outputs and occasionally conflicted with user inputs, resulting in unstable flight behavior. Simulated latency average 290us
- **Diagnosed:** Early model outputs revealed limitations in the design and data flow between high and low layers, indicated the need for an improved model and larger training datasets.
- **Future:** Currently redesigning the pipeline with a focus on low-level data processing, system integration, and control structures to support more dynamic acrobatic maneuvers.
- **Summarize:** Demonstrated the feasibility of the overall control pipeline. Further refinement of both the AI and robotics architectures is needed.

Internship_Work.md

Project: Cross-Platform ITAM Endpoint Device Agent

- **Goal:** Develop a lightweight, memory-safe IT asset monitoring daemon for Linux and Windows.
- **Main components:** Rust, Tokio, Unix Domain Sockets, Min-Heap scheduler, flamegraph, hyperfine.
- **Principle:** Architecture is decoupled into discrete binaries (launcher, worker, fetcher) linked by IPC. This isolates faults-a fetcher panic will not crash the main daemon. Precise task scheduling is managed via a custom Min-Heap scheduler.
- **Execution:** Built the multi-binary Rust agent handling OS-specific system calls. Implemented the IPC layer for data passing and profiled memory/CPU bottlenecks using flamegraphs and hyperfine.
- **Results:** Successfully delivered a functioning cross-platform agent capable of scheduled data fetching and resilient inter-process communication without memory leaks.
- **Summarize:** Delivered a foundational MVP, successfully demonstrating Rust's utility as a highly effective choice for building memory-safe, low-overhead system architectures.

Capstone_Project_Plan.md

Project: VTOL Acrobatic flight enhancement and Obstacle Avoidance with RL

Goals: Design an AI-driven control pipeline that explores and exploits the highly non-linear flight envelope of VTOL while maintaining deterministic spatial safety.

Main components: C++, ROS2 (Jazzy), PyTorch (RL), Voxblox (ESDF), Gazebo Harmonic, PX4 Autopilot.

Principle: Designed an AI-enhanced control system architecture, aimed at providing expert-level control capabilities to pilots of all skill levels. The Acrobatic AI input include control intent (human pilot or planer), drone state (estimated by PX4) and a spatial extraction from the Voxblox ESDF map, latent by a 3D CNN. RL training happen progressively from Hover → Tracking → Obstacle avoidance, each task with it own reward and an entropy objective to explore the flight envelope. The output are normalized actuator control signal, sent to PX4 mixer to perform action. There a deterministic Reactive fallback with simple APF path planning for extra safety.

Summarize: Proposing a novel, pilot intent driven AI architecture designed to maximize VTOL flight capability while maintaining a collision safety.